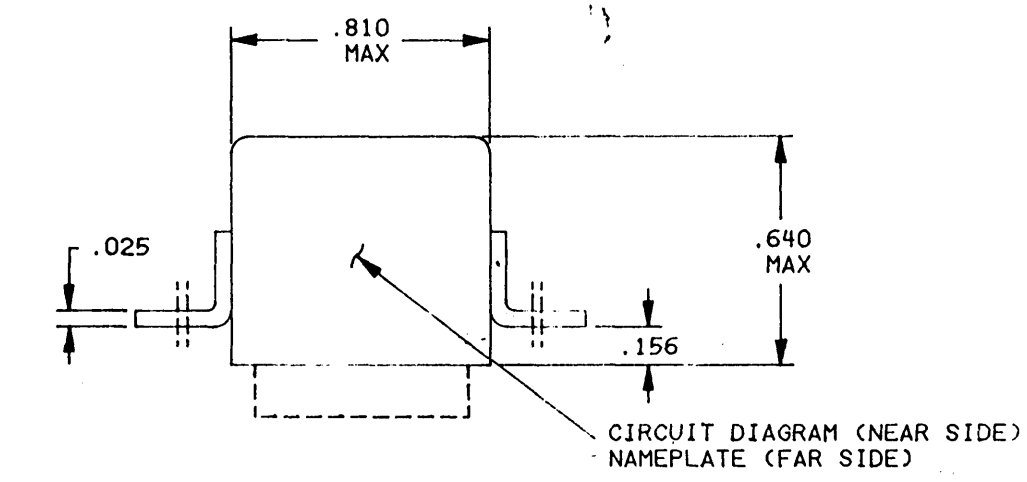
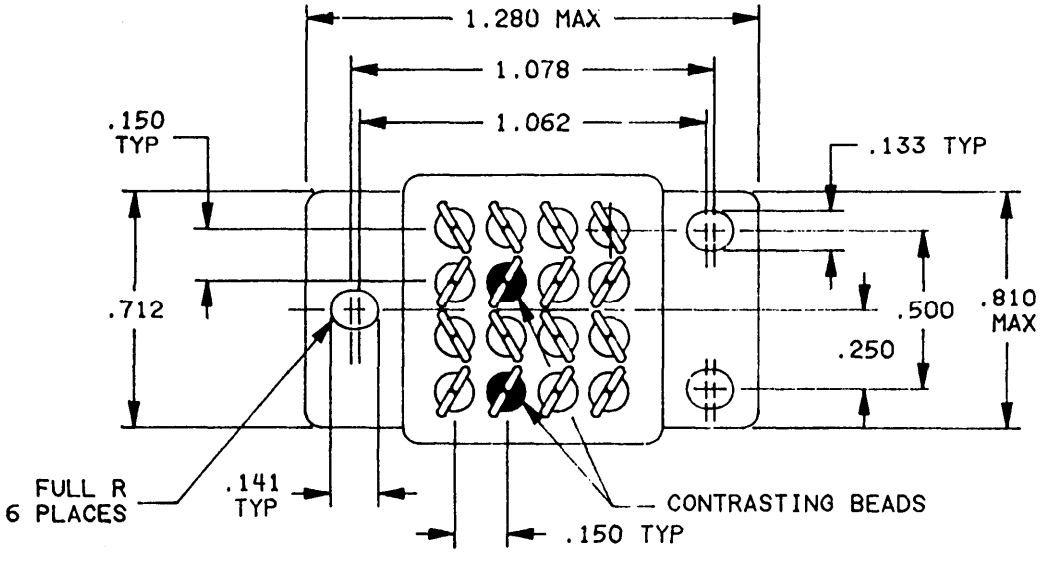
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC (TYPE I), MAGNETIC LATCH (WITH INTERNAL DIODES FOR COIL TRANSIENT SUPPRESSION), PERMANENT MAGNET DRIVE, LOW LEVEL TO 5 AMPERES 4PDT, ALL WELDED, HERMETICALLY SEALED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-5106.



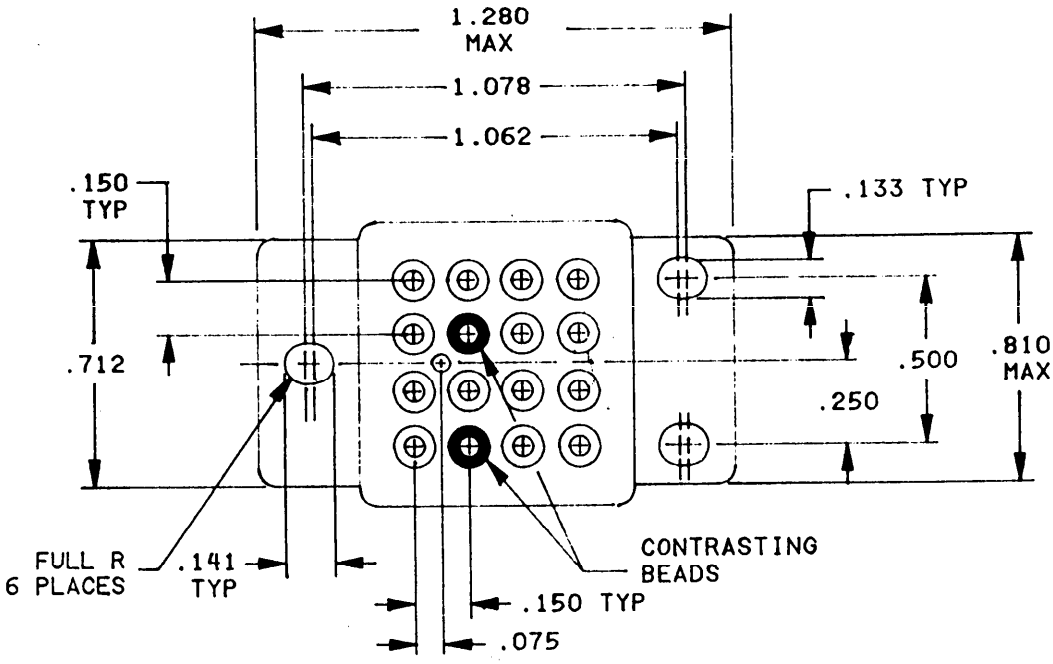


CONFIGURATION A

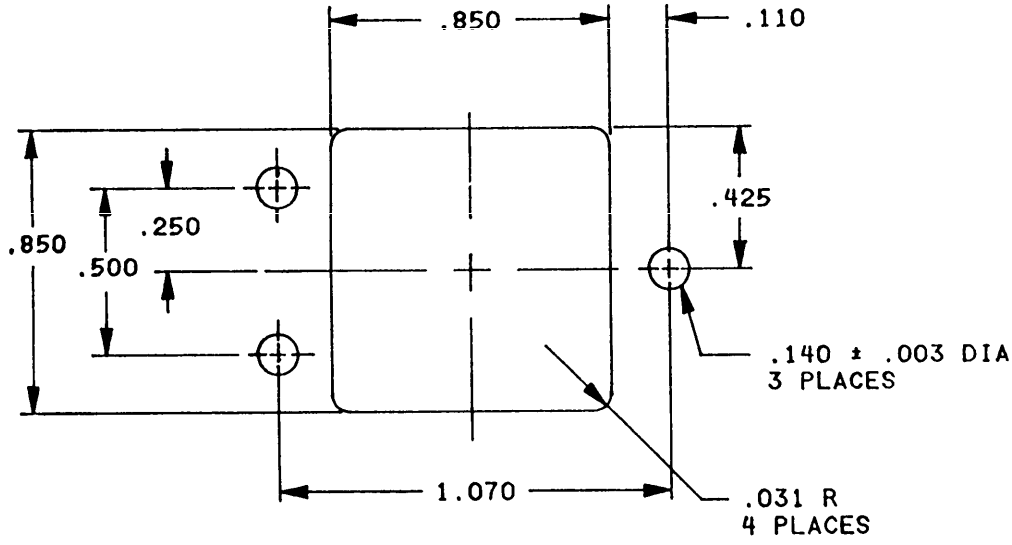
FIGURE 1. Outline drawing.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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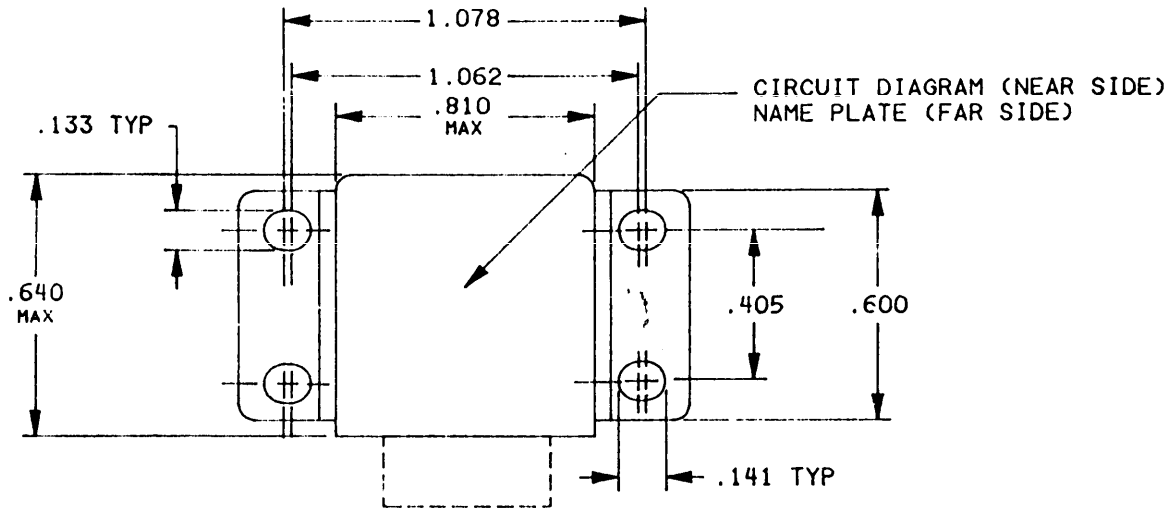
Configuration A

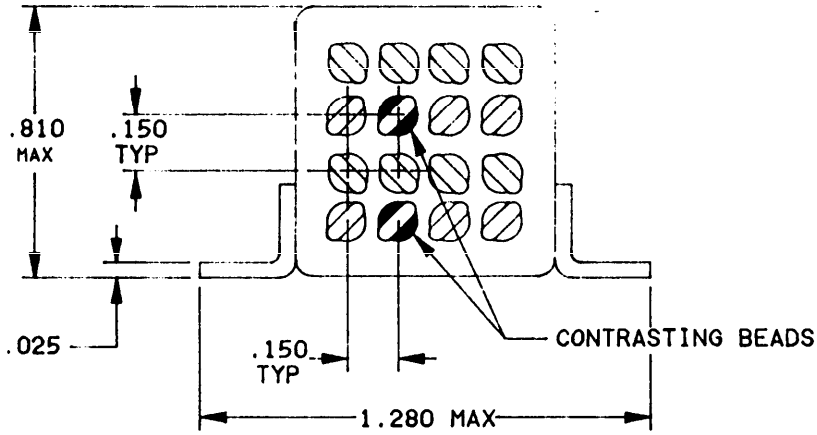


MOUNTING HOLE LAYOUT

FIGURE 1. Outline drawing - Continued.

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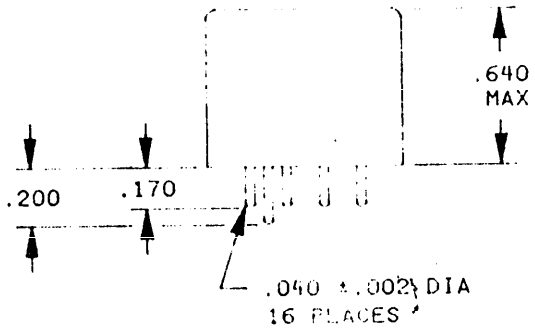


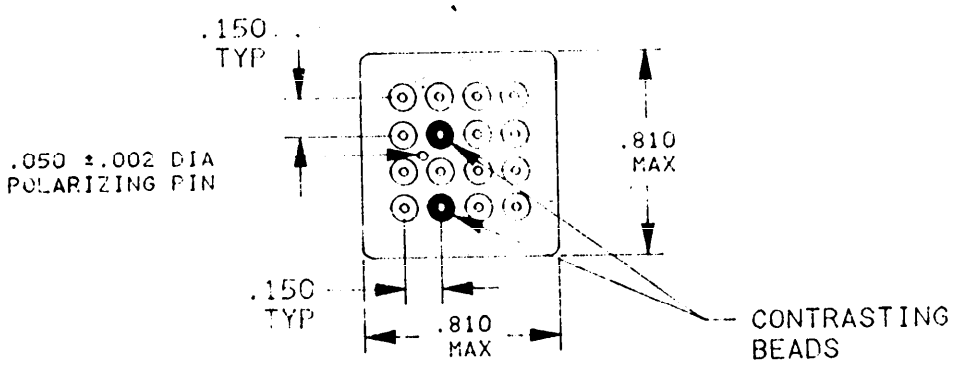


Configuration B

FIGURE 1. Outline drawing - Continued.

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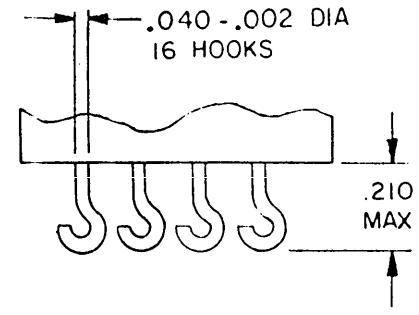




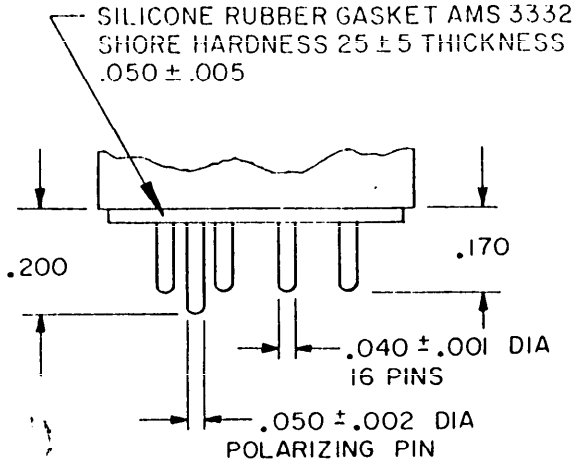
Configuration C

FIGURE 1. Outline drawing - Continued.

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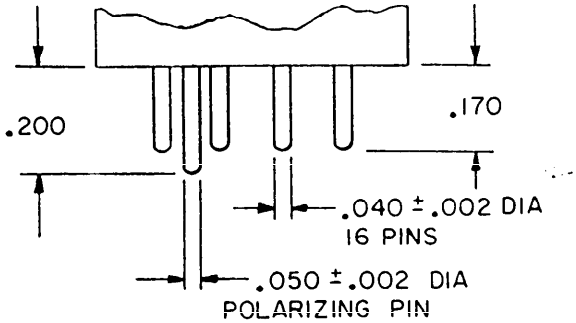


SOLDER HOOK PIN – TIN PLATED

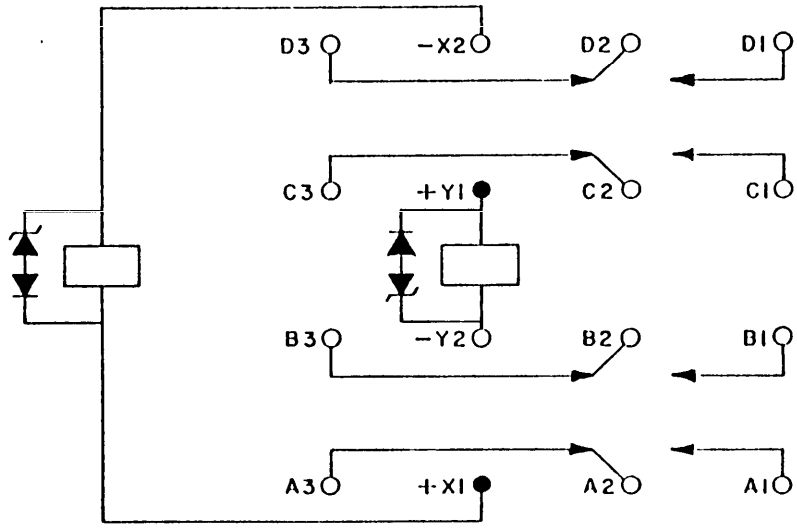


SOCKET PIN

GOLD PLATED PER MIL-G-45204,
TYPE II, CLASS 1, UNDERPLATING
NICKEL, 50 TO 150 MICROINCHES THICK



SOLDER PIN – TIN PLATED



CIRCUIT DIAGRAM

TO CLOSE NO. 1 CONTACTS (OPERATE)
ENERGIZE X1 AND X2

TO CLOSE NO. 3 CONTACTS (RESET)
ENERGIZE Y1 AND Y2

FIGURE 1. Outline drawing - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.140	3.56	.640	16.26
.002	0.05	.141	3.58	.712	18.08
.003	0.08	.150	3.81	.810	20.57
.025	0.64	.156	3.96	.850	21.59
.031	0.79	.170	4.32	1.062	26.97
.040	1.02	.200	5.08	1.070	21.18
.050	1.27	.210	5.33	1.078	27.38
.075	1.91	.250	6.35	1.280	32.51
.110	2.79	.425	10.80		
.133	3.38	.500	12.70		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (25.4 mm).
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized Bistable.
6. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electro-magnet flux.

FIGURE 1. Outline drawing - Continued.

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REQUIREMENTS:

Coil data: See table I.

Duty rating: Continuous.

Operational data: See tables I and II.

Time current characteristics: See table III. 1/

Physical data:

Dimensions and configuration: See figure 1.

Weight (maximum): .064 pound, 29.1 grams.

Terminal type: See figure 1 and table I.

Finish: Paint optional.

ENVIRONMENTAL CHARACTERISTICS:

Temperature rating: -70°C to +125°C.

Maximum altitude: 300,000 feet.

Shock G-level: 200 g.

Duration: 6 ms.

Maximum duration contact opening: 10 µs.

Vibration, sinusoidal:

G-level: 30 g.

Frequency range: 70 to 3,000 Hz.

Vibration, random:

Applicable standard: MIL-STD-202, method 214, 0.4 g^2/Hz, test condition: IG.

Duration: 15 minutes each plane.

Acceleration: 15 g.

ELECTRICAL CHARACTERISTICS:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Time-current relay characteristics: See table III.

1/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence as listed in table III. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

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TABLE I. Operating characteristics.

Dash no. M6106/53	Coil data: Coils 1/	Nominal: Volts	Nominal: Frequency Hz	Nominal: Resistance *10% at 25°C 2/	Maximum: Volts	Maximum: Amperes at 25°C	Maximum pickup voltage: Normal 3/	Maximum pickup voltage: High temperature test	Maximum pickup voltage: Continuing current test	Time (milliseconds, maximum): Operate	Time (milliseconds, maximum): Reset	Contact bounce: Main NO	Contact bounce: Main NC	Contact bounce: Auxiliary NO	Contact bounce: Auxiliary NC	Terminal type	Mating socket	Configuration
-001 -002 4/	X1, X2, Y1, Y2	28	DC	600	29	.054	18	19.8	22.5	6	6	.5	.5	---	---	Solder hook	N/A	A
-003 -004 4/	X1, X2, Y1, Y2	28	DC	600	29	.054	18	19.8	22.5	6	6	.5	.5	---	---	Solder pin	[illegible]	A
-005 [illegible] 4/	X1, X2, Y1, Y2	28	DC	600	29	.054	18	19.8	22.5	6	6	.5	.5	---	---	Solder pin	N/A	A
[illegible]	X1, X2, Y1, Y2	28	DC	600	29	.054	18	19.8	22.5	6	6	.5	.5	---	---	Solder hook	N/A	B
[illegible]	X1, X2, Y1, Y2	28	DC	600	29	.054	18	19.8	22.5	6	6	.5	.5	---	---	Solder pin	N/A	C
[illegible] 4/	X1, X2, Y1, Y2	28	DC	600	29	.054	18	19.8	22.5	6	6	.5	.5	---	---	Solder pin	N/A	B

1/ Back EMF 42 V dc maximum.
2/ Coil resistance not directly measurable at coil terminals.
3/ Over the temperature range.
4/ Screened relays shall be tested to the group A established reliability requirements only and additional life testing is not required. For group A inspection, operational reliability shall be performed at the low level run in.

TABLE II. Rated contact load (amperes per pole) case grounded.

Type of load	Endurance (life) operating cycles x 10^3	28 V dc Main NO	28 V dc Main NC	115 V ac, 1 phase Main 400 Hz	115/200 V ac, 3 phase Main 400 Hz	See appropriate notes
Resistive	100	5	5	5	5	
Inductive	100	---	---	---	---	
Inductive	20	3	3	5	5	
Motor	100	2	2	3	3	
Lamp	100	1	1	1	1	
Transfer load	---	---	---	---	---	1/
Mechanical life reduced current	400	1.25	1.25	1.25	1.25	
Intermediate current	50	Applicable per specification				
Low level	100	Applicable per specification				2/ 3/
Mixed loads	As applicable	Applicable per specification				4/

1/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.
2/ For low level, the following shall apply:
   a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10- to 50-millivolt open-circuit voltage, 100-ohm maximum contact resistance.
   b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.
3/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.
4/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

MIL-R-6106/53(AT)

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Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case:	500 V rms	350 V rms
Coil to coil:	500 V rms	350 V rms
All other points:	1,000 V rms	750 V rms

Dielectric withstanding voltage (altitude) 2/:

	80,000 ft	300,000 ft
Coil to case:	250 V rms	500 V rms
Coil to coil:	250 V rms	500 V rms
All other points:	250 V rms	500 V rms

Maximum contact drop, initial: 0.150 volt.

After life tests: 0.175 volt.

Overload current: 20 amperes dc, 30 amperes ac.

Rupture current: 25 amperes dc, 40 amperes ac.

Part number: M6106/53-(dash number from table I).

TABLE III. Time-current relay characteristics at 25°C.

1	10 A	1 hour
2	15 A	20 seconds
3	37.5 A	2 seconds
4	75 A	.53 second

CAUTION: Compare with time-current characteristics of associated circuit protective device.

Qualification by similarity: See table IV.

2/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

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ABLE IV. Qualification by similarity.

Part number M6106/53	Load A (dc)	Dynamics				Environmental A (dc)
		A 1/	B	C	D	
-001	1, 2, 3		1			1, 2, 3
-002	1, 2, 3		1			1, 2, 3
-003	1, 2, 3	1				1, 2, 3 2/
-004	1, 2, 3	1				1, 2, 3 2/
-005	1, 2, 3		1			1, 2, 3
-006	1, 2, 3		1			1, 2, 3
-007	1, 2, 3			1		1, 2, 3
-008	1, 2, 3			1		1, 2, 3
-009	1, 2, 3				1	1, 2, 3
-010	1, 2, 3				1	1, , 3
-011	1, 2, 3			1		1, 2, 3
-012	1, 2, 3			1		1, 2, 3

1/ Terminals: Socket pin type terminals are considered worst case. Solder pins and hooks may be qualified by similarity.

2/ Socket pins (with gasket seals), are considered nonsimilar to solder pins and solder hooks. Each type must be tested.

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Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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